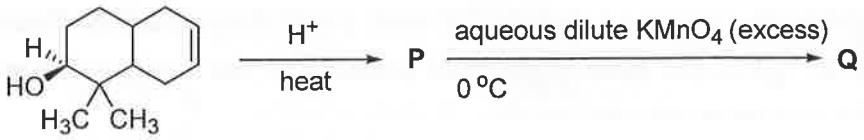
PART II : CHEMISTRY

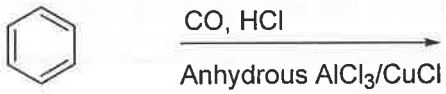
SECTION 1 (Maximum Marks: 32)

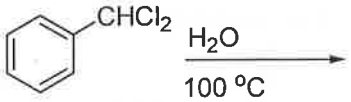
- This section contains **EIGHT** questions
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive
- For each question, darken the bubble corresponding to the correct integer in the ORS
- Marking scheme:
  - +4 If the bubble corresponding to the answer is darkened
  - 0 In all other cases

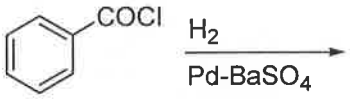
Q.21 The number of hydroxyl group(s) in **Q** is

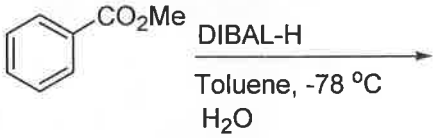


Q.22 Among the following, the number of reaction(s) that produce(s) benzaldehyde is

I. 

II. 

III. 

IV. 

Q.23 In the complex acetylbromidodicarbonylbis(triethylphosphine)iron(II), the number of Fe–C bond(s) is

Answers for the above questions

Ans for Q.21: 4	**Ans for Q.22: 4**	**Ans for Q.23: 3**

Q.24 Among the complex ions, $[Co(NH_2\text{-}CH_2\text{-}CH_2\text{-}NH_2)_2Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, $[Fe(H_2O)_4(OH)_2]^+$, $[Fe(NH_3)_2(CN)_4]^-$, $[Co(NH_2\text{-}CH_2\text{-}CH_2\text{-}NH_2)_2(NH_3)Cl]^{2+}$ and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) *cis-trans* isomerism is

Q.25 Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is

Q.26 The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.10 M). If $\lambda^0_{X^-} \approx \lambda^0_{Y^-}$, the difference in their pK_a values, $pK_a(HX) - pK_a(HY)$, is (consider degree of ionization of both acids to be $\ll 1$)

Q.27 A closed vessel with rigid walls contains 1 mol of $^{238}_{92}U$ and 1 mol of air at 298 K. Considering complete decay of $^{238}_{92}U$ to $^{206}_{82}Pb$, the ratio of the final pressure to the initial pressure of the system at 298 K is

Q.28 In dilute aqueous H_2SO_4, the complex diaquodioxalatoferrate(II) is oxidized by MnO_4^-. For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is

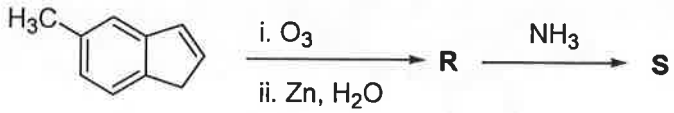
Answers for the above questions

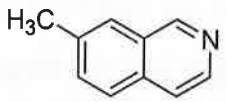
Ans for Q.24: 6	**Ans for Q.25: 6**	**Ans for Q.26: 3**
Ans for Q.27: 9	**Ans for Q.28: 8**	

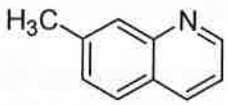
SECTION 2 (Maximum Marks: 32)

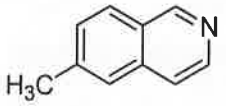
- This section contains **EIGHT** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are)correct
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme:
  - +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
  - 0 If none of the bubbles is darkened
  - –2 In all other cases

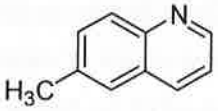
Q.29 In the following reactions, the product **S** is



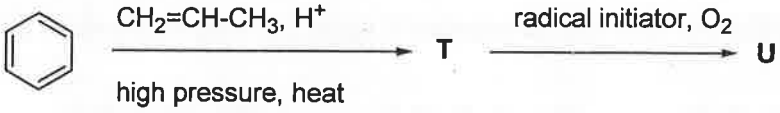
(A) 

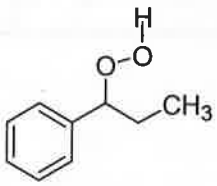
(B) 

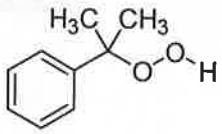
(C) 

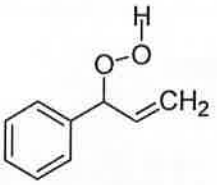
(D) 

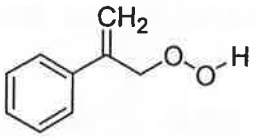
Q.30 The major product **U** in the following reactions is



(A) 

(B) 

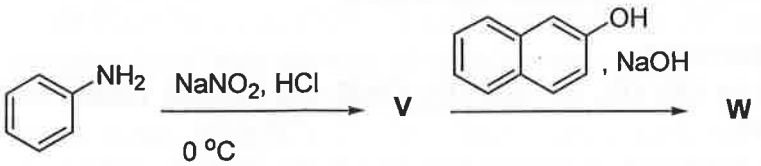
(C) 

(D) 

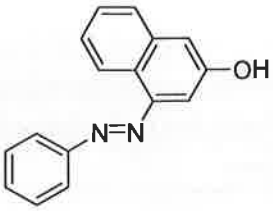
Answers for the above questions

Ans for Q.29: (A)	Ans for Q.30: (B)

Q.31 In the following reactions, the major product W is



(A) 

(B) 

(C) 

(D) 

Q.32 The correct statement(s) regarding, (i) $HClO$, (ii) $HClO_2$, (iii) $HClO_3$ and (iv) $HClO_4$, is(are)

(A) The number of Cl=O bonds in (ii) and (iii) together is two

(B) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three

(C) The hybridization of Cl in (iv) is sp^3

(D) Amongst (i) to (iv), the strongest acid is (i)

Q.33 The pair(s) of ions where BOTH the ions are precipitated upon passing H_2S gas in presence of dilute HCl, is(are)

(A) Ba^{2+}, Zn^{2+} (B) Bi^{3+}, Fe^{3+} (C) Cu^{2+}, Pb^{2+} (D) Hg^{2+}, Bi^{3+}

Answers for the above questions

Ans for Q.31:	Ans for Q.32:	Ans for Q.33:
(A)	(B) and (C)	(C) and (D)

Q.34 Under hydrolytic conditions, the compounds used for preparation of linear polymer and for chain termination, respectively, are

(A) CH_3SiCl_3 and $Si(CH_3)_4$ (B) $(CH_3)_2SiCl_2$ and $(CH_3)_3SiCl$

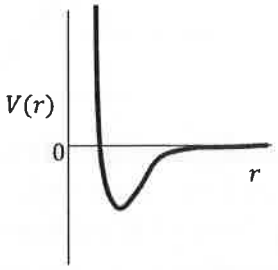
(C) $(CH_3)_2SiCl_2$ and CH_3SiCl_3 (D) $SiCl_4$ and $(CH_3)_3SiCl$

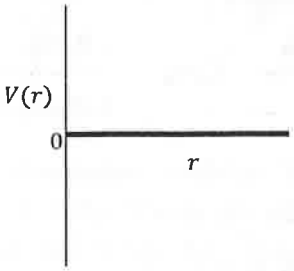
Q.35 When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2. The **TRUE** statement(s) regarding this adsorption is(are)

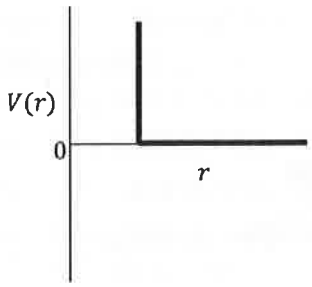
(A) O_2 is physisorbed (B) heat is released

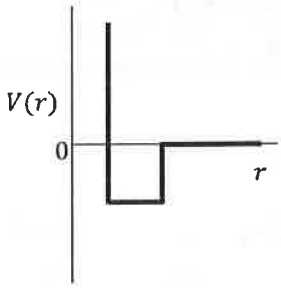
(C) occupancy of π^*_{2p} of O_2 is increased (D) bond length of O_2 is increased

Q.36 One mole of a monoatomic real gas satisfies the equation $p(V-b)=RT$ where b is a constant. The relationship of interatomic potential $V(r)$ and interatomic distance r for the gas is given by

(A)


(B)


(C)


(D)


Answers for the above questions

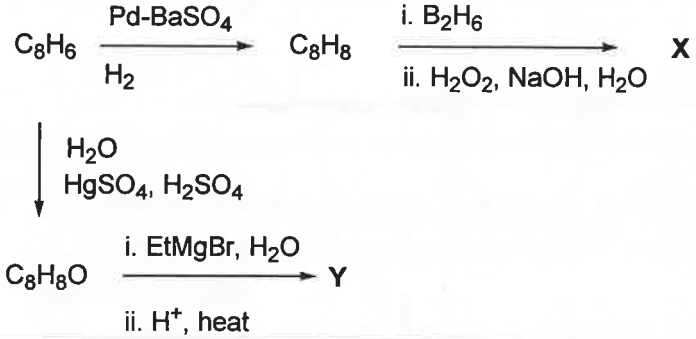
Ans for Q.34:	Ans for Q.35:	Ans for Q.36:
(B)	(B), (C) and (D)	(C)

SECTION 3 (Maximum Marks: 16)

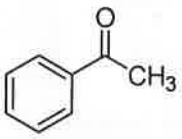
- This section contains **TWO** paragraphs
- Based on each paragraph, there will be **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- Marking scheme:
  - +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened
  - 0 If none of the bubbles is darkened
  - –2 In all other cases

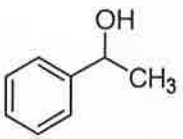
PARAGRAPH 1

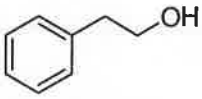
In the following reactions

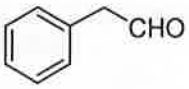


Q.37 Compound **X** is

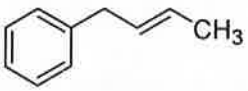
(A) 

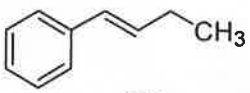
(B) 

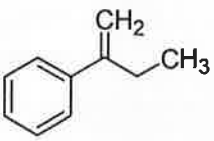
(C) 

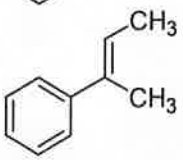
(D) 

Q.38 The major compound **Y** is

(A) 

(B) 

(C) 

(D) 

Answers for the above questions

Ans for Q.37: (C)	Ans for Q.38: (D)

PARAGRAPH 2

When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.7 °C was measured for the beaker and its contents (**Expt. 1**). Because the enthalpy of neutralization of a strong acid with a strong base is a constant (–57.0 kJ mol^{-1}), this experiment could be used to measure the calorimeter constant.

In a second experiment (**Expt. 2**), 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.0 M NaOH (under identical conditions to **Expt. 1**) where a temperature rise of 5.6 °C was measured.

(Consider heat capacity of all solutions as 4.2 J g^{-1} K^{-1} and density of all solutions as 1.0 g mL^{-1})

Q.39 Enthalpy of dissociation (in kJ mol^{-1}) of acetic acid obtained from the **Expt. 2** is

(A) 1.0 (B) 10.0 (C) 24.5 (D) 51.4

Q.40 The pH of the solution after **Expt. 2** is

(A) 2.8 (B) 4.7 (C) 5.0 (D) 7.0

END OF PART II : CHEMISTRY

Answers for the above questions

Ans for Q.39: (A)	Ans for Q.40: (B)